

Associations of Contemporary Screen Time Modalities With Early Adolescent Nutrition



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ABSTRACT

OBJECTIVE: To determine the associations between screen time across several contemporary screen modalities (eg, television, video games, text, video chat, social media) and adherence to the Mediterranean-DASH (dietary approaches to stop hypertension) intervention for neurodegenerative delay (MIND) diet in early adolescents.

METHODS: We analyzed data from the Adolescent Brain Cognitive Development study of 9 to 12-year-old adolescents in the United States. Multiple linear regression analyses examined the relationship between self-reported screen time measures at baseline (year 0) and the 1-year follow-up (year 1) and caregiver-reported nutrition assessments at year 1, providing a prospective and cross-sectional analysis. Cross-sectional marginal predicted probabilities were calculated.

RESULTS: In a sample of 8267 adolescents (49.0% female, 56.9% white), mean age 10 years, total screen time increased from 3.80 h/d at year 0 to 4.61 h/d at year 1. Change in total screen time

from year 0 and year 1 was associated with lower nutrition scores at year 1. *Prospective:* Screen time spent on television, video games, and videos at year 0 was associated with lower nutrition scores at year 1. *Cross-sectional:* Screen time spent on television, video games, videos, texting, and social media at year 1 was associated with lower MIND diet scores at year 1.

CONCLUSIONS: Both traditional (television) and several contemporary modalities of screen time are associated, prospectively and cross-sectionally, with lower overall diet quality, measured by the MIND diet nutrition score, in early adolescents. Future studies should further explore the effect of rising digital platforms and media on overall adolescent nutrition.

KEYWORDS: adolescent; Mediterranean-dietary approaches to stop hypertension intervention for neurodegenerative delay diet; nutrition; screen time

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WHAT'S NEW

In a demographically diverse, sample of early adolescents in the United States, total screen time, television, YouTube videos, and video games were associated with a lower MIND diet nutrition score, prospectively and cross-sectionally; texting and social media were associated cross-sectionally.

Digital media has become a central part in the lives of contemporary adolescents. The variety of digital media devices for leisure continues to increase as concern over the potential harm of this time spent on screens also rises. On average, adolescents spend 4.2 h/d on their screens.¹ From a developmental perspective, early adolescence typically

includes a decrease in parental monitoring, supervision, and influence, coupled with an increase in time spent with peers, including time spent online.² Higher screen time has been shown to have negative consequences on adolescents' physiological and psychological well-being through sedentary behavior, poor sleep quality, and obesity.³

One growing area of interest is the relationship between prolonged screen time and poor nutrition. Adolescence is characterized by a transition from being fed by parents to more autonomy about an adolescent's own food intake.⁴ Previous studies have shown the link between screen time and unhealthy eating behaviors among adolescents, such as skipping breakfast, the consumption of sugary drinks, sweets, and fast food, and a decreased intake of vegetables and fruits.^{5,6} However, literature has traditionally focused on

television-watching as an exposure, linking increased time viewing to a higher frequency of unhealthy dietary behaviors.^{7,8} Few studies have illustrated this relationship with video games, associating increased screen time with increased consumption of nutritionally deficient food and poor adherence to the Mediterranean diet,⁹ a widely acknowledged healthy diet.^{10,11}

The amount of time spent on television-watching and video games has now been eclipsed by the amount of screen time spent on electronic devices among adolescents.¹ Some studies have demonstrated that digital platforms—YouTube, Instagram, Facebook—have become common spaces for adolescents to engage with unhealthy food advertisements, which have been associated with higher consumption of these foods.^{12,13} A cross-sectional study in Korea showed that both duration and content type of smartphone use were associated with dietary risk factors (eg, more sugar-sweetened beverage intake, less frequent intakes of fruits and vegetables) among adolescents.¹⁴

However, the prospective and cross-sectional associations between contemporary screen modalities (eg, texting, video chatting, and social media) and diet quality have yet to be examined in the United States. The effects of each may be nuanced, depending on factors such as engagement and level of interaction.¹³ There is also a paucity of data regarding screen time and the Mediterranean-DASH (dietary approaches to stop hypertension) intervention for neurodegenerative delay (MIND) diet in adolescents, which has shown both cardiometabolic and neuroprotective effects in older adults.¹⁵ The recommendations of the MIND diet differ slightly from those of the commonly measured Healthy Eating Index-2015, which is based on the USDA Dietary Guidelines for Americans, by emphasizing specific foods that promote brain health such as leafy greens and berries.¹⁶ In fact, research in pre-adolescents has shown that adherence to the MIND diet, and not Healthy Eating Index-2015, was associated with increased performance on a task assessing attentional inhibition.¹⁷ Our study also distinguishes itself through utilizing the MIND Diet score as a comprehensive measure of diet quality, in contrast to existing literature that predominantly categorizes nutritional patterns related to screen time based on the consumption of specific food items individually (sweets, chips, fruits, vegetables).^{5–7,10,14,18,19}

The objective of this study was to determine the prospective and cross-sectional associations between contemporary forms of screen time and diet quality, corresponding to the MIND diet, among a socio-demographically diverse, population-based sample of 9 to 12-year-old early adolescents in the United States.

METHODS

This study presents a prospective and cross-sectional analysis of data from the Adolescent Brain Cognitive Development (ABCD) study, an ongoing longitudinal study of adolescent health and cognitive development in 11,962 children recruited from 21 sites around the United States.²⁰ Recruitment and sampling of participants predominantly occurred through educational institutions, with careful consideration given to

factors such as gender, race and ethnicity, socioeconomic status, and urbanicity to mitigate potential biases in sample selection.²¹ Specifically, the ABCD study employed a primary method of probability sampling of US schools across the 21 catchment areas. These areas were distributed across the nation's 4 major regions (Northeast, South, Midwest, West) and exhibited demographic and socioeconomic diversity. The study utilized annual databases from the National Center for Education Statistics, providing sociodemographic profiles of students in each public and private school within the catchment areas. Employing a stratified, probability sampling approach at the school level for each of the 21 sites minimized systematic biases in recruitment. Consequently, the sample is epidemiologically informed, although it may not comprehensively mirror the US population due to voluntary research participation and self-selection biases.

This investigation used information from the ABCD 4.0 release for the baseline (2016–2018, 9–10 years old) and 1-year follow-up (2017–2019, 10–11 years old) assessments.²⁰ Detailed information about the study has been published elsewhere.²⁰ Participants' missing data for the Youth Screen Time Survey, Child Nutrition Assessment, and demographics were excluded ($n = 3695$), leaving a final sample of 8267 adolescents ([Supplementary Appendix A](#)). Institutional review board approval was received from the University of California, San Diego, and the individual recruitment sites. Written informed consent was obtained from parents, while written assent was obtained from participants.

INDEPENDENT VARIABLES

The ABCD Youth Screen Time Survey was recorded at the baseline and year-1 follow-up assessments, providing data on participants' self-reported screen time.²² Participants reported the number of hours per day they spent on either viewing/streaming television shows or movies, watching/streaming videos (eg, YouTube), playing video games, texting, video chatting (eg, Skype, Facetime), and social media (eg, Facebook, Instagram, Twitter) separately for weekdays and weekend days. We performed a weighted average calculation of the participants' typical weekday and weekend screen time consumption to obtain a typical week measure ($[\text{weekday average } 5] + [\text{weekend average } 2])/7$.²² This single measure allowed us to incorporate an estimated average daily screen use over 1 week to allow for a single screen time outcome. We reported the weighted total screen time average as a continuous variable and as a categorical variable (in 2 h/d increments) after obtaining this total for each screen time modality used by participants. We also calculated a variable that represents the change in total screen time (ie, total screen time at year 1 subtracted from total screen time at year 0).

DEPENDENT VARIABLES

Given that adolescents have a tendency to underreport dietary energy intake and are prone to errors when completing food records,²³ the nutritional status data are reported by parents, who have been shown to be valid reporters for their children's diet.²⁴ Parents completed a Child Nutrition

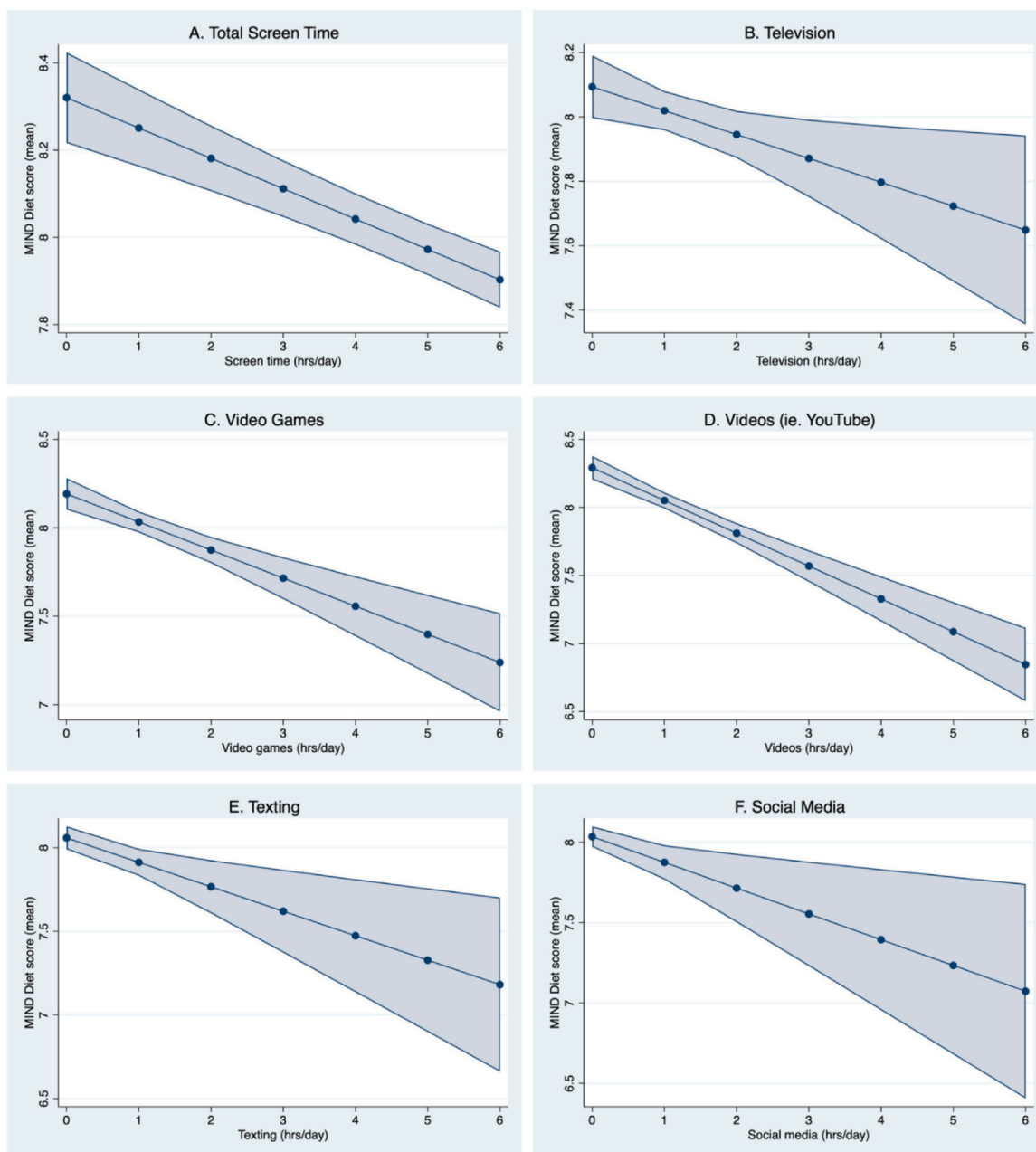


Figure. Cross-sectional adjusted predicted scores with 95% confidence intervals of screen time and MIND diet score. Modalities that were significant ($P < .05$) in the cross-sectional linear regressions: (A) total screen time and MIND diet score; (B) television and MIND diet score; (C) video games and MIND diet score; (D) videos and MIND diet score; (E) texting and MIND diet score; (F) social media and MIND diet score. MIND indicates Mediterranean-dietary approaches to stop hypertension intervention for neurodegenerative delay.

Assessment at the 1-year follow-up, which was modeled after the MIND diet questionnaire²⁵—a 14-item Food Frequency Questionnaire that assesses intake of food in protective groups (eg, whole grains, vegetables, berries, nuts, poultry, beans, fish, and olive oil) and neurodegenerative groups (eg, fried foods, pastries and sweets, butter, and cheese). The ABCD study omitted the measure of wine consumption in the MIND diet, which is 15-items in other literature.²⁵ The Child Nutrition Assessment asks about the child's intake of a certain serving size of each food group over the course of a week. For each participant survey, the total sum score was calculated, which was used as a measure of overall nutritional status.

COVARIATES

Participants' parents reported sociodemographic characteristics of their children at baseline, such as participants' sex at birth (male or female), race and ethnicity (Asian, Black, Latino, Native American, White, or other), annual household income (less than \$25 K, \$25 K–\$50 K, \$50 K–\$75 K, \$75 K–\$100 K, \$100 K–\$200 K, and greater than \$200 K), highest parent education level (college education or more, or high school education or less), and age. Depressive symptoms (Child Behavior Checklist depressive problems T-score)^{26,27} and parental monitoring (Parent Monitoring Scale)^{28–30} at baseline were also included as potential confounders for the association between screen time and nutrition.

STATISTICAL ANALYSIS

Data analyses for this study were performed in 2023 using Stata 15.1. Multivariable linear regressions were used to investigate the associations between screen time use, total screen time (continuous and categorical) and each of the 6 modalities (continuous), at baseline (year 0) and 1-year follow-up (year 1), and adherence to the MIND diet at the 1-year follow-up (year 1) for a total of 16 individual regression models. We also examined a separate model that included baseline total screen time and changes in screen time (independent variables) with MIND diet score (dependent variable). Covariates, measured at year 0, were adjusted for (itemized in the section above). Marginal predicted probabilities were computed following cross-sectional linear regressions to get standardized estimates to aid with the interpretation of the findings (Figure). Propensity weights were applied to match key sociodemographic variables in the ABCD study to the American Community Survey from the US Census.³¹

RESULTS

The analytic sample ($n = 8267$) was approximately matched by sex (49% female), was racially and ethnically diverse (43.1% non-White), and, on average was 10 years old (Table 1). Total screen time increased from 3.80 h/d at baseline (year 0) to 4.61 h/d at the 1-year follow-up (year 1). For both years (years 0 and 1), the screen modalities with the greatest time reported were television (1.27 h/d) followed by videos (1.00–1.22 h/d) and video games (1.00–1.21 h/d). The average MIND diet nutrition sum score at year 1 was 8 points out of 14.

Table 2 shows results from the adjusted linear regression models examining the prospective and cross-sectional associations between screen time use at year 0 and year 1 and adherence to the MIND diet at year 1. Each additional hour per day of total screen time at both year 0 (−0.07, 95% CI: −0.09, −0.05) and year 1 (−0.07, 95% CI: −0.09, −0.05) was associated with a lower nutrition score at year 1. Each additional hour per day of screen time at year 0 and year 1 in the form of watching videos (−0.18, 95% CI: −0.24, −0.12); [−0.24, 95% CI: −0.30, −0.19]), video games (−0.15, 95% CI: −0.21, −0.09]; [−0.16, 95% CI: −0.22, −0.10]), and television (−0.14, 95% CI: −0.20, −0.07]; [−0.07, 95% CI: −0.14, −0.01]) was associated with a lower nutrition score at year 1. Texting (−0.15, 95% CI: −0.24, −0.05) and social media (−0.16, 95% CI: −0.28, −0.05) screen use at year 1 were associated with lower nutrition scores, cross-sectionally.

There was generally a dose-response association with greater total screen time categories and lower MIND diet score (Table 2). Compared to 0 to 2 h/d of baseline total screen time, > 2 to 4 (−0.31, 95% CI: −0.45, −0.17), > 4 to 6 (−0.50, 95% CI: −0.68, −0.32), and > 6 (−0.65, 95% CI: −0.84, −0.45) hours per day of total screen time were associated with lower MIND diet scores at year 1.

Predicted probabilities computed after cross-sectional linear regressions of screen time and nutrition score

Table 1. Sociodemographic and Screen Time Characteristics of 8267 Adolescent Brain Cognitive Development Study Participants

Sociodemographic Characteristics (Baseline)	Mean (SD)/%
Age at baseline (years)	9.94 (0.63)
Sex (%)	
Female	49.0%
Male	50.1%
Race and ethnicity (%)	
White	56.9%
Latino/Hispanic	19.2%
Black	14.3%
Asian	5.4%
Native American	3.0%
Other	1.3%
Household income (%)	
Less than \$25,000	15.3%
\$25,000–\$49,999	18.1%
\$50,000–\$74,999	15.7%
\$75,000–\$99,999	14.8%
\$100,000–\$199,999	26.9%
\$200,000 and greater	9.4%
Parent education level (%)	
Parent with college education or more	89.7%
Parent with less than high school	10.3%
Parent monitoring sum score	4.38 (0.51)
Depressive symptoms	53.8 (5.92)
MIND diet sum score	8.00 (2.45)
Screen time variables (hours per day, baseline)	
Total screen time	3.80 (3.02)
Television shows/movies	1.27 (1.03)
Videos (eg, YouTube)	1.00 (1.15)
Video games	1.00 (1.09)
Texting	0.23 (0.53)
Video chat	0.19 (0.47)
Social media	0.11 (0.39)
Total screen time categories (%)	
0–2	33.5%
> 2–4	31.5%
> 4–6	16.6%
> 6	18.4%
Screen time variables (hours per day, year 1)	
Total screen time	4.61 (3.57)
Television shows/movies	1.27 (1.06)
Videos (eg, YouTube)	1.22 (1.21)
Video games	1.21 (1.21)
Texting	0.41 (0.73)
Video chat	0.28 (0.63)
Social media	0.23 (0.60)
Total screen time categories (%)	
0–2	26.1%
> 2–4	28.8%
> 4–6	17.7%
> 6	27.4%
Change in screen time from baseline to year 1 (hours per day)	0.82 (3.21)

MIND indicates Mediterranean-dietary approaches to stop hypertension intervention for neurodegenerative delay; SD, standard deviation.

Adolescent Brain Cognitive Development propensity weights were applied based on the American Community Survey from the US Census.

(Figure) reveal that across a visual examination of the figures, higher values of screen time are associated with lower nutrition scores.

Table 2. Associations of Screen Time (Year 0 and Year 1) and MIND Diet Score (Year 1) in the Adolescent Brain Cognitive Development Study (N = 8267)

	Prospective Analysis		Cross-Sectional Analysis	
	Screen Time (year 0) and MIND Diet Score		Screen Time (year 1) and MIND Diet Score	
	B (95% CI)	P	B (95% CI)	P
Screen time (continuous, hours per day)				
Total screen time	-0.07 (-0.09, -0.05)	<.01	-0.07 (-0.09, -0.05)	<.01
Television	-0.14 (-0.20, -0.07)	<.01	-0.07 (-0.14, -0.01)	.02
Video games	-0.15 (-0.21, -0.09)	<.01	-0.16 (-0.22, -0.10)	<.01
Videos (YouTube)	-0.18 (-0.24, -0.12)	<.01	-0.24 (-0.30, -0.19)	<.01
Texting	-0.09 (-0.22, 0.04)	.17	-0.15 (-0.24, -0.05)	.01
Video chat	0.05 (-0.10, 0.19)	.53	-0.09 (-0.19, 0.02)	.10
Social media	-0.12 (-0.29, 0.05)	.17	-0.16 (-0.28, -0.05)	.01
Total screen time categories (hours per day)				
0–2	reference	—	reference	—
> 2–4	-0.31 (-0.45, -0.17)	<.01	-0.27 (-0.41, -0.12)	<.01
> 4–6	-0.50 (-0.68, -0.32)	<.01	-0.47 (-0.65, -0.29)	<.01
> 6	-0.65 (-0.84, -0.45)	<.01	-0.72 (-0.90, -0.54)	<.01

MIND indicates Mediterranean-dietary approaches to stop hypertension intervention for neurodegenerative delay.

Bold indicates $P < .05$. The B coefficient in the cells represents the abbreviated output from a series of linear regression models with nutrition sum score as the dependent variable and screen time as the independent variable. Thus, the table represents the outputs from 16 regression models in total. Adolescent Brain Cognitive Development propensity weights were applied based on the American Community Survey from the US Census. Covariates: race/ethnicity, sex, age, household income, parent education, depression, parent monitoring, and study site.

Table 3. Associations of Baseline Screen Time, Change in Screen Time (Baseline to Year 1), and MIND Diet Score (Year 1) in the Adolescent Brain Cognitive Development Study (N = 8267)

	MIND Diet Score	
	B (95% CI)	P
Total screen time, baseline (h/d)	-0.09 (-0.12, -0.07)	<.01
Change in total screen time, baseline to year 1 (h/d)	-0.05 (-0.08, -0.03)	<.01

MIND indicates Mediterranean-dietary approaches to stop hypertension intervention for neurodegenerative delay.

Bold indicates $P < .05$. The B coefficient in the cells represents the abbreviated output from a linear regression model with nutrition sum score as the dependent variable and baseline screen time and change in screen time (from baseline to year 1) as the independent variables. Adolescent Brain Cognitive Development propensity weights were applied based on the American Community Survey from the US Census. Covariates: race/ethnicity, sex, age, household income, parent education, depression, parent monitoring, and study site.

Increases in total screen time from baseline to year 1 were associated with lower MIND diet scores, even after accounting for baseline total screen time (Table 3).

DISCUSSION

In this demographically diverse sample of 9 to 12-year-old early adolescents in the United States, we found that greater total screen time use was associated with lower MIND diet nutrition score, prospectively and cross-sectionally. From baseline to year 1, total screen time use increased by approximately 1 h/d, demonstrating an upward trend. Television was associated with lower adolescent nutrition scores at year 1, prospectively and cross-sectionally, consistent with prior studies linking TV time to poor diet quality.^{7,8,14} However, extending prior work, the present study demonstrated significant decrements in diet quality in relation to

contemporary modalities of media use. Time spent playing video games and watching online videos at year 0 and year 1 were associated with lower nutrition scores at year 1. It is notable that videos and video games had a stronger negative association with the MIND diet score than the more extensively researched television exposure. More time spent texting and on social media were also associated with a lower nutrition score cross-sectionally, with stronger associations than television.

Our findings are consistent with prior literature that outlined the associations between time spent watching television, playing video games, and smartphone use with poor diet quality among adolescents.^{7,8,14} One prior study of Norwegian early adolescents found that an hour increase in television viewing time was associated with less fruit (-0.12 times per week) and vegetable (-0.19 times per week) consumption but greater soft drink (0.13 times per week) and snack (0.07 times per week) consumption.⁸ Building on prior studies on this topic that have established the relationship between more screen time and unhealthy or “junk” foods, here we used a comprehensive questionnaire to demonstrate that more screen time was predictive of overall poorer diet quality. Our study showed that contemporary forms of screen time (eg, videos, texting, and social media on smartphones) may have stronger associations with poor diet quality than television-watching alone.

The link between screen time and poor adolescent nutrition may be explained by several different mechanisms. Increased time on screens can increase exposure to unhealthy eating practices through online platforms or television advertisements, which may lead adolescents to turn to these nutritionally deficient food options.¹³ Additionally, for all ages, there is an increase in unhealthy snacking behaviors and overall food consumption when

watching screens.³² Time spent on screens may also reduce healthy lifestyle behaviors such as time spent eating a balanced meal, exercising, and sleeping.^{33,34} Additional research should examine the policy implications of food marketing to kids and interventions to promote sleep hygiene.

There are limitations to consider. Although we did include a prospective analysis, the observational nature of our study precludes establishing causality, and the relationship between screen time and nutrition could be bidirectional. The absence of a baseline nutrition score limited our analysis. The effect sizes observed in our study were relatively small; however, the decrements in diet quality were expressed relative to 1 hour increase in screen time, whereas total screen time was 3.80 at baseline and 4.61 h at year 1. As shown by the large SD around these means, some youth are engaging in vastly more screen time. Some screen time estimates could be double reported (eg, if an adolescent double reported texting and social media if the texting occurred within a social media app); however, this would downward bias associations between total screen time and nutrition toward the null hypothesis. Our reliance on self-report measures also introduced the potential for measurement biases.

Strengths of the study include the inclusion of contemporary screen modalities and the use of the MIND diet Food Frequency Questionnaire. Additionally, we leveraged data from a large, national, demographically diverse prospective cohort with a 1-year follow-up, enhancing the generalizability of our findings. Studies in aging populations, for which MIND was designed, consistently show that a difference in MIND diet score, as small as 1 point, is related to significant cognitive decline, performance, and memory complaints.³⁵ Nevertheless, it is unknown how MIND diet scores in youth impact future risk.

CONCLUSION

This study has clinical relevance and application for pediatricians caring for early adolescents. Clinicians should assess for nutritional status at each pediatric health assessment visit, which includes a nutritional history and assessment of food intake and behavior.³⁶ Clinicians should also consider assessing for screen time and provide resources for parents such as a Family Media Use plan,³⁷ which could involve screen-free times during family meals to encourage communication within families and to avoid eating while distracted in front of screens. Along with the rise of social media use, there have been significant efforts to utilize digital media for nutrition interventions among adolescents^{38,39}; however, public health professionals should be aware of the overall negative associations between contemporary modes of screen time identified in this study and nutrition when designing such interventions. Moreover, the US Surgeon General issued an advisory on youth social media and mental health in May 2020.⁴⁰ The association of social media and poor nutrition is 1 potential mechanism that

could explain links between social media and mental health. Future studies should explore potential pathways through which social media may be associated with nutrition, and how parental and social interactions may affect these relationships in adolescents.

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The funders had no role in the study analysis, decision to publish the study, or the preparation of the manuscript.

DECLARATION OF COMPETING INTEREST

The authors have no conflicts of interest to disclose.

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SUPPLEMENTARY DATA

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.acap.2024.01.023](https://doi.org/10.1016/j.acap.2024.01.023).

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